

EXPERIMENT 3

POLYALPHABETIC CIPHER (VIGENÈRE CIPHER)

AIM

To implement the Vigenère (Polyalphabetic) Cipher substitution technique using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Read the keyword from the user.
3. Repeat the keyword until its length matches the plaintext.
4. Encrypt each character by adding the positions of the plaintext character and the corresponding key character modulo 26.
5. Display the encrypted (cipher) text.
6. Decrypt the cipher text by subtracting the key character value from the cipher character modulo 26.
7. Display the decrypted text.

PYTHON CODE

```
def generate_key(text, key):  
    key = key.upper()  
    text = text.upper()  
    new_key = ""  
    j = 0  
  
    for ch in text:  
        if ch.isalpha():  
            new_key += key[j % len(key)]  
            j += 1  
        else:  
            new_key += ch  
    return new_key
```

```
def encrypt(text, key):  
    text = text.upper()  
    key = generate_key(text, key)  
    cipher = ""  
  
    for i in range(len(text)):  
        if text[i].isalpha():  
            x = (ord(text[i]) - ord('A') + ord(key[i]) - ord('A')) % 26  
            cipher += chr(x + ord('A'))  
        else:  
            cipher += text[i]  
  
    return cipher
```

```
def decrypt(cipher, key):  
    cipher = cipher.upper()  
    key = generate_key(cipher, key)  
    plain = ""  
  
    for i in range(len(cipher)):  
        if cipher[i].isalpha():  
            x = (ord(cipher[i]) - ord(key[i])) % 26  
            plain += chr(x + ord('A'))  
        else:  
            plain += cipher[i]
```

```
return plain
```

```
plaintext = input("Enter Plain Text: ")
```

```
key = input("Enter Key: ")
```

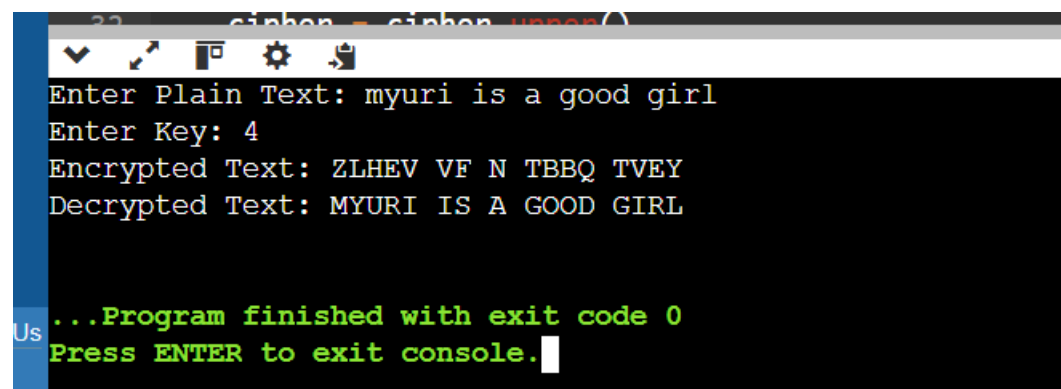
```
cipher = encrypt(plaintext, key)
```

```
print("Encrypted Text:", cipher)
```

```
plain = decrypt(cipher, key)
```

```
print("Decrypted Text:", plain)
```

OUTPUT



```
?? cipher = cipher_upper()
Enter Plain Text: myuri is a good girl
Enter Key: 4
Encrypted Text: ZLHEV VF N TBBQ TVEY
Decrypted Text: MYURI IS A GOOD GIRL
...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT

Thus, the Vigenère (Polyalphabetic) Cipher substitution technique was implemented successfully using Python.